

Teaching Statement

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My teaching and mentoring prepares students to develop and research software by *learning their tools*. This approach to instruction teaches students to use all facets of every utility available to them when solving problems, empowering them to not only create robust software, but to do so with alacrity. This produces a sense of mastery, initiating a positive feedback loop in which students perpetually seek opportunities to expand their knowledge further. To provide students with such opportunities, I recruit them to conduct research with me on software engineering and security, giving them the chance to hone their skills while contributing to the broader field of Computer Science.

Teaching Techniques and Research Mentoring Methods

I Work Backwards from Concrete Objectives

Before preparing any course material, I choose specific student learning outcomes. This provides a concrete course structure, and enables students to check their progress towards course goals. I also use this tactic when collaborating with undergraduate researchers, beginning new research projects by first designing research questions. This helps students remember overarching research objectives when performing research tasks.

Teaching example. To teach students about version control with Git, I worked backwards by first choosing the specific Git commands I wanted students to learn (e.g., add, commit, push). I then provided [illustrated examples](#) of these commands during lecture, and tested student understanding by assigning [a project](#) on using Git to create a local repository, stage and commit changes to it, and finally upload those changes to a remote repository. The class average for this project has always been an A, attesting to the efficacy of working backwards to design course material.

Mentoring example. I began working with a team of undergraduate research assistants to conduct research on securing software build phases by preparing and disseminating [a list of the project's goals](#). I then decomposed these goals down into specific tasks, and created GitHub issues with instructions for undergraduate assistants on how to complete them. Utilizing GitHub issues had the additional benefit of enabling all members of the team to track our progress towards meeting the project's goals. After a semester of organized collaboration, this work has resulted in a publication under review at the 49th IEEE/ACM International Conference on Software Engineering.

I Leverage the Familiar

I start with familiar content before moving into unknown territory. This not only engages students, but also facilitates learning by easing students into new information. This technique also helps me on-board research assistants onto projects by first giving them small and easy tasks before assigning more complex ones.

Teaching example. To teach students how to solve software engineering problems by slowly decomposing larger problems into smaller ones, I begin with the familiar fable of the tortoise and the hare. Students already know the fable, and so the message becomes intuitive: expedience leads to failure, whereas prudence begets success. Establishing this message early eases my way in to giving students specific examples of careful programming practices.

Mentoring example. When I recruited Rasmin Ahmned, a student in my COP 3402 course section, to assist me with my research on securing software build systems, I first assigned her a task that built off what she had learned in my class. Specifically, I tasked her with cataloging the commands to configure, compile, and test projects using GNU Make, a tool which she had already been using to complete projects for COP 3402. Since she was already familiar with Make, she was able to complete the swiftly, and has since used her research experience to obtain an in-person interview for an internship at Amazon.

I Reward Success

To incentivize students to be continuous learners, I allow students to continuously resubmit late projects for a small point penalty, and offer extra credit for implementing bonus features in programming assignments. This approach teaches students to persevere past their failures, rather than to fear their mistakes. In a research setting, this approach encourages students to obtain novel results by exploring new ideas.

Teaching example. When teaching COP 3402, I test student ability to create systems software by tasking them with implementing their own shell in C. Many students find this task difficult, because it requires using in conjunction all the Linux system calls they will have learned up that point in the course. Consequently, many students initially score only partial points on this assignment, which in another course would be their final grade for the project. But, since I allow students to resubmit the project until the end of the semester to regain lost points, many students keep working to improve their scores. By the end of the semester, the average class grade for the project is always an A. But more importantly, continuously working on the project helps students reinforce their knowledge of the Linux shell, with the median score for shell-related questions on the final exam for COP 3402 consistently being at least 85%.

Mentoring example. When conducting research on translating C preprocessor macros to C language constructs, I gave my undergraduate assistant [Joseph Zalusky](#) great freedom in choosing how to implement a tool for performing macro-to-C translation. As his mentor, I supported Joey by supplying him with the formal rules for performing the translation, giving him code review, and providing general guidance on how to conduct research. With our combined efforts we produced the first semantically-aware macro-to-C conversion tool, [MerC](#), which has led to a conditionally-accepted paper at the 41st IEEE/ACM International Conference on Automated Software Engineering. For his hard work Joey was made a co-author on the paper, and he used his research experience to obtain a full-time software engineering position at Moog, Inc.

Summary of Teaching Experience

While pursuing my PhD at the University of Central Florida, I taught **COP-3402: Systems Software** for both the [Spring 2026](#) and [Summer 2026](#) semesters. I began teaching the course with a curriculum prepared by my advisor, and I have been gradually updating this curriculum in response to student suggestions. I have also used student feedback to expand my teaching arsenal with new teaching techniques, such as in-lecture questions, in-class examples, and [live lecture recordings](#). For these efforts I have earned high course (4.73 / 5 average) and instructor (4.62 / 5 average) evaluation ratings.

Before teaching COP 3402, I served as GTA for the course, with my advisor being the instructor, during both the Spring and Fall of 2025. During these semesters I led small (30 students or fewer) labs, in which I would answer student questions, review project assignments, and provide enriching examples of real-world applications of course content. The effectiveness of these techniques is evinced by the glowing reviews left by students, not only for myself ([4.93 / 5](#)), but also for my advisor, who, with my assistance as a GTA, earned the highest ratings he had ever received for the course's structure (4.92 / 5) and instruction (4.89 / 5). On top of positive student reviews, I have also received GTA excellence awards from UCF at both the [college](#) and [university](#) level. This latter award was only given to one out of 800+ GTAs across all of UCF, and I was the sole recipient.

Summary of Mentoring Experience

While completing my PhD at the University of Central Florida, I have mentored [many undergraduate students](#) across several different research projects. In 2024 I recruited [Zachary Burkett](#) and [Joseph Zalusky](#) and to assist me with research on the semantic-aware translation of C preprocessor macros to C programming language constructs. This collaboration resulted in a conditionally-accepted paper at the 41st IEEE/ACM International Conference on Automated Software Engineering, with both undergraduate students being co-authors. Both these students leveraged their research experience to secure positions they were seeking, with Joey obtaining a full-time software engineering role at Moog Inc., and Zachary being accepted to the Masters of Sound and Music Computing program at the Pompeu Fabra University.

In 2026 I recruited a team of seven undergraduate students to assist me with research on securing software build phases against pipeline poisoning attacks: Adam Betinsky, [David Gusmao](#), Michael Johnson, Brayden

Coggin, Daniel Henriquez, Thiago Airoidi, and Rasmin Ahmed. Four of these students discovered an interest in research while taking a course for which I served in a teaching capacity, with Adam, Michael, and David being enrolled in lab sessions I led as a GTA for COP 3402 in 2025, and Rasmin being a student in the section of COP 3402 I taught in 2026. The work this team of students has produced has led to a publication currently under review at the 49th IEEE/ACM International Conference on Software Engineering. As with my previous undergraduate collaborations, these students have used the research experience they gained while working with me to obtain competitively-sought after roles, with Adam securing an internship at Keystone Strategy, and Rasmin obtaining an in-person interview for an internship at Amazon.

Conclusion

My teaching and mentoring center on equipping students with the skills to become proficient software engineers. I achieve this through three key techniques: **working backwards**, **leveraging the familiar**, and **rewarding success**. Using these techniques, I have achieved high course (4.73 / 5 average) and instructor (4.62 / 5 average) evaluation ratings, as well multiple awards for GTA excellence. I have also applied these three techniques to undergraduate mentoring to produce several top-tier publications with undergraduate researchers, many of whom have leveraged the research experience they gained working with me to obtain positions in industry and academia. I will continue to improve on these techniques to deliver quality instruction to my students, and enriching research opportunities to my mentees.